Workshop 2: Cross-validation and model selection

TECH6

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See GitHub repository for notebooks and data:

https://github.com/richardfoltyn/TECH6-H26

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Exercise 1: Selecting and evaluating a polynomial model

Consider the model

$$y_i = f(x_i) + \epsilon_i, \qquad f(x) = \sin(2\pi(x + 0.1)),$$

where x_i is uniform on $[0, 1]$ and the independently and identically distributed (i.i.d.) errors satisfy $\epsilon_i \overset{\text{iid}}{\sim} N(0, 0.5^2)$. For this exercise, assume that we don't know the true model and are instead approximating f with a polynomial of degree D, i.e.,

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_D x_i^D + u_i,$$

where u_i is the residual not explained by the polynomial.

In this exercise, we use cross-validation (CV) to select a polynomial degree, refit the selected model, and evaluate its predictions on independent observations. We first compute the validation scores for each degree, then reproduce the same selection using higher-level scikit-learn functions.

Setup

The true function, sample generator, and plotting functions are provided in the following cells. We use 100 training observations for all model selection and fitting. We generate an independent test sample only after selecting and fitting the models.

[1]:
```
import matplotlib.pyplot as plt
import numpy as np
from numpy.random import default_rng
```

[2]:
```
def fcn(x):
    """
    Evaluate the true sine relationship.

    Parameters
    ----------
    x : array-like
        Input values

    Returns
    -------
    array
        True function values
    """
    return np.sin(2 * np.pi * (x + 0.1))
```

[3]:
```
def generate_sample(n_obs, sigma, rng=None):
    """
    Generate a sample with uniform inputs and normally distributed errors.

    Parameters
    ----------
    n_obs : int
        Number of observations
    sigma : float
        Standard deviation of the errors
    rng : generator, optional
        Random number generator used for the draws

    Returns
    -------
    X : array
        Feature matrix with one column
    y : array
        Observed outcomes
    """
    if rng is None:
        rng = default_rng(seed=1234)

    # Draw uniform inputs
    x = rng.uniform(size=n_obs)
    # Draw normally distributed errors
    epsilon = rng.normal(scale=sigma, size=n_obs)
    # Add the errors to the true function values
    y = fcn(x) + epsilon
    # Reshape x to a feature matrix with one column
    X = x[:, None]
    return X, y
```

[4]:
```
def plot_sample(X, y):
    """
    Plot the training observations and true function.

    Parameters
    ----------
    X : array
        Feature matrix with one column
    y : array
        Observed outcomes
    """
    x_grid = np.linspace(0, 1, 301)
    _fig, ax = plt.subplots(figsize=(4.5, 3), layout='constrained')
```

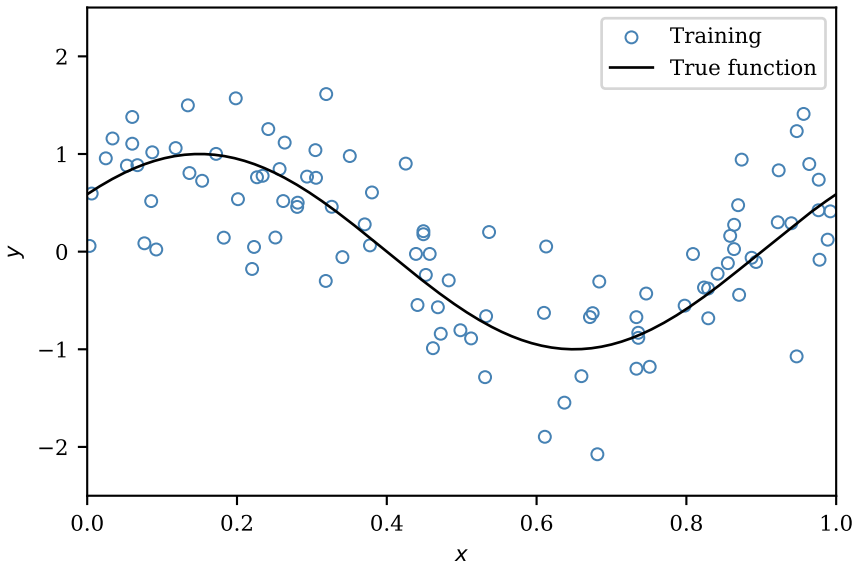
```
    ax.scatter(X[:, 0], y, s=20, c='none', ec='steelblue', lw=0.75, label='Training')
    ax.plot(x_grid, fcn(x_grid), c='black', lw=1, label='True function')
    ax.set(xlabel='$x$', ylabel='$y$', xlim=(0, 1), ylim=(-2.5, 2.5))
    ax.legend()
    return ax
```

[5]:
```
# Number of observations
n_obs = 100

# Standard deviation of the errors
sigma = 0.5

# Generate the sample
X, y = generate_sample(n_obs, sigma)

# Plot the sample
_ = plot_sample(X, y)
```



Part 1 — Constructing the pipeline

As a first step, build a `Pipeline` called `pipe` with these named steps:

1. `input_scale`: use `StandardScaler` to center and scale the input.
2. `poly`: use `PolynomialFeatures` with `degree=1` and `include_bias=False`.
3. `scale`: standardize the expanded features with another `StandardScaler`.
4. `lr`: fit `LinearRegression` with `fit_intercept=True`.

Explain why the scalers must be inside the pipeline when we use cross-validation.

Test that the pipeline works by fitting it to the sample generated above.

[6]:
```
from sklearn.linear_model import LinearRegression
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import PolynomialFeatures, StandardScaler

# TODO: Build pipe with the specified steps

# TODO: Test pipeline by fitting it on X and y.
```

Solution.

We scale the input before taking powers to improve numerical stability, then scale the expanded columns. Both scalers are fitted on each fold's training observations. Fitting them before CV would let the held-out observations influence the transformations.

Excluding the polynomial bias column avoids a redundant constant column when the regression includes an intercept.

[7]:
```
from sklearn.linear_model import LinearRegression
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import PolynomialFeatures, StandardScaler

pipe = Pipeline(
    steps=[
        ('input_scale', StandardScaler()),
        ('poly', PolynomialFeatures(degree=1, include_bias=False)),
        ('scale', StandardScaler()),
        ('lr', LinearRegression(fit_intercept=True)),
    ]
)

# Fit the pipeline to the sample
pipe.fit(X, y)
```

[7]:
```
Pipeline(steps=[('input_scale', StandardScaler()),
                ('poly', PolynomialFeatures(degree=1, include_bias=False)),
                ('scale', StandardScaler()), ('lr', LinearRegression())])
```

Part 2 — Computing MSEs for one degree

In this part, you are asked to implement a function that returns the average training and validation MSEs across CV folds for a given polynomial degree.

1. Define a CV splitter using `KFold` with ten shuffled folds and a fixed seed of 1234.
2. Complete the function `compute_average_mse()` below. Use `cross_validate()` to compute the training and validation mean squared errors (MSEs) for each fold. The function takes an estimator, the training data, and a CV splitter, and returns the mean training and validation MSEs across folds.

   Hint: Pass `scoring='neg_mean_squared_error'` and `return_train_score=True` as arguments to `cross_validate()`.

   Hint: Don't forget to negate the scores before returning the MSEs.

Test your function implementation with the pipeline from Part 1, the CV splitter from (1), and the training data generated earlier.

[8]:
```
from sklearn.model_selection import cross_validate


def compute_average_mse(estimator, X, y, cv):
    """
    Compute mean training and validation MSEs across CV folds.

    Parameters
    ----------
    estimator : estimator
        Regression model or pipeline
```

```
    X : array-like
        Explanatory variables
    y : array-like
        Response variable
    cv : splitter
        Cross-validation splitter

    Returns
    -------
    mse_train : float
        Mean training MSE across folds
    mse_validate : float
        Mean validation MSE across folds
    """
    # TODO: Run cross_validate() with training scores and negative MSE scoring.

    # TODO: Negate and average the training and validation scores.

    # TODO: Return the two mean MSEs.


# TODO: Define a CV splitter with ten shuffled folds and a random state of 1234.

# TODO: Test compute_average_mse() with the pipeline, sample, and CV splitter.
```

Solution.

We first define the CV splitter using the `KFold` class:

[9]:
```
from sklearn.model_selection import KFold

# Define the CV splitter
cv_splitter = KFold(n_splits=10, shuffle=True, random_state=1234)
```

Next, we use `cross_validate()` in our implementation of `compute_average_mse()` to fit a fresh copy of the pipeline on each fold's training observations. We average the returned scores across folds and negate them.

[10]:
```
from sklearn.model_selection import cross_validate


def compute_average_mse(estimator, X, y, cv):
    """
    Compute mean training and validation MSEs across CV folds.

    Parameters
    ----------
    estimator : estimator
        Regression model or pipeline
    X : array-like
        Explanatory variables
    y : array-like
        Response variable
    cv : splitter
        Cross-validation splitter

    Returns
    -------
    mse_train : float
        Mean training MSE across folds
    mse_validate : float
        Mean validation MSE across folds
```

```
    """
    scores = cross_validate(
        estimator,
        X,
        y,
        cv=cv,
        scoring='neg_mean_squared_error',
        return_train_score=True,
    )

    # Negate scores to get MSEs and average across folds
    mse_train = -scores['train_score'].mean()
    mse_validate = -scores['test_score'].mean()
    return mse_train, mse_validate
```

We can test the function with the pipeline and CV splitter defined above:

[11]:
```
# Test with the pipeline and CV splitter defined above
mse_train, mse_validate = compute_average_mse(pipe, X, y, cv_splitter)

print(f'Training MSE: {mse_train:.4f}')
print(f'Validation MSE: {mse_validate:.4f}')
```

```
Training MSE: 0.5246
Validation MSE: 0.5482
```

Part 3 — Plotting the validation curve

We now want to use the function `compute_average_mse()` implemented in Part 2 to compute the mean training and validation MSEs for polynomial degrees 1 through 15. We are therefore "manually" computing the validation curve. Recall that the validation curve plots the training and validation scores as a function of a hyperparameter, in this case the polynomial degree.

1. Loop over degrees 1 through 15. For each degree, you can either recreate the pipeline with the new degree or use `pipe.set_params()` to change `poly__degree` on an existing pipeline (see code template below).

   Call your `compute_average_mse()` function with the same CV splitter defined in Part 2 to compute the mean training and validation MSEs for each degree and store these.

2. Find and report the degree with the smallest mean validation MSE.

3. Use the supplied `plot_validation_curve()` function to plot the training and validation MSEs as a function of the polynomial degree.

[12]:
```
def plot_validation_curve(degrees, mse_train, mse_validation, best_degree):
    """
    Plot mean training and validation MSEs against polynomial degree.

    Parameters
    ----------
    degrees : array-like
        Candidate polynomial degrees
    mse_train : array-like
        Mean training MSE for each degree
    mse_validation : array-like
        Mean validation MSE for each degree
    best_degree : int
        Degree with the smallest mean validation MSE
    """
    _fig, ax = plt.subplots(figsize=(5.5, 3), layout='constrained')
```

```
    ax.plot(degrees, mse_train, c='black', marker='x', ms=4, label='Training')
    ax.plot(
        degrees, mse_validation, c='steelblue', marker='o', ms=4, label='Validation'
    )
    ax.axvline(best_degree, ls=':', lw=0.75, c='black')
    ax.set(xlabel='Polynomial degree', ylabel='Mean CV MSE', xticks=degrees)
    ax.legend()
```

[13]:
```
# Degrees of the polynomial features to consider
degrees = np.arange(1, 16)

for degree in degrees:
    # Update the degree of the polynomial features in the pipeline
    pipe.set_params(poly__degree=degree)

    # TODO: Compute the mean training and validation MSEs for the current degree

    # TODO: Store the mean training and validation MSEs for each degree

# TODO: Find and report the degree with the smallest mean validation MSE

# TODO: Plot the validation curve using the supplied `plot_validation_curve()` function
```

Solution.

[14]:
```
# Degrees of the polynomial features to consider
degrees = np.arange(1, 16)

# Lists to store the mean training and validation MSEs for each degree
mse_train_mean = []
mse_validate_mean = []

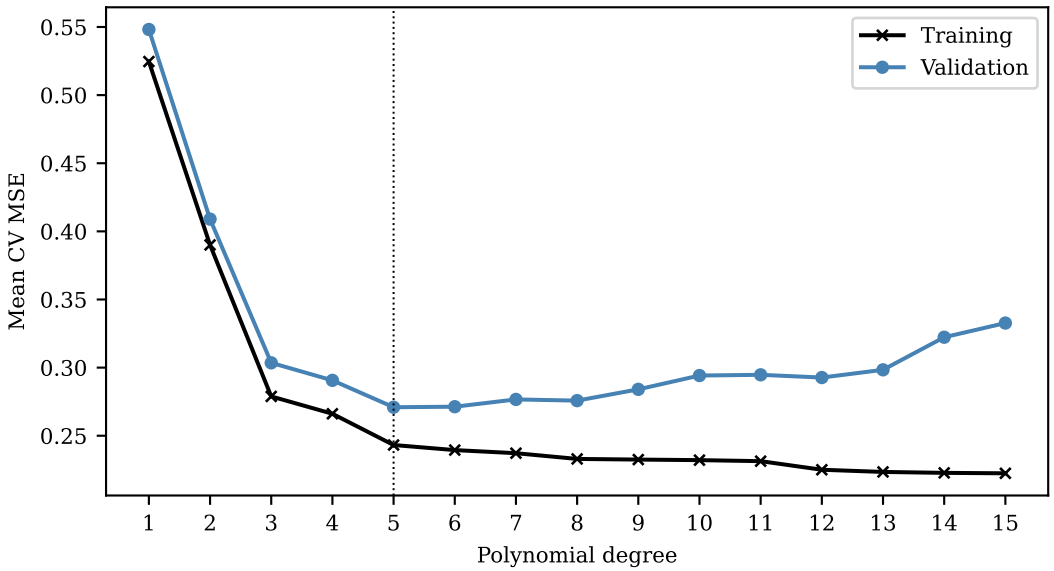
for d in degrees:
    # Update the degree of the polynomial features in the pipeline
    pipe.set_params(poly__degree=d)
    # Compute the mean training and validation MSEs for the current degree
    mse_train_d, mse_validate_d = compute_average_mse(pipe, X, y, cv_splitter)

    # Store the mean training and validation MSEs for current degree
    mse_train_mean.append(mse_train_d)
    mse_validate_mean.append(mse_validate_d)

# Find index of the degree with the smallest mean validation MSE
imin = np.argmin(mse_validate_mean)
# Recover the degree with the smallest mean validation MSE
dmin = degrees[imin]
print(f'Degree with minimum validation MSE: {dmin}')
print(f'Minimum CV MSE: {mse_validate_mean[imin]:.3f}')
```

```
Degree with minimum validation MSE: 5
Minimum CV MSE: 0.271
```

[15]:
```
# Plot validation curve and indicate the optimal degree
plot_validation_curve(degrees, mse_train_mean, mse_validate_mean, dmin)
```



Degree 5 minimizes the mean validation MSE at about 0.271. Training MSE falls as the polynomial becomes more flexible, while validation MSE eventually rises. The validation curve need not be a smooth U: its values fluctuate across neighboring degrees.

Part 4 — Using `validation_curve()`

Instead of computing the validation curve manually, we can use the convenience function `validation_curve()`.

1. Repeat the degree comparison using `validation_curve()`. Use the same pipeline, candidate degrees, CV splitter, and negative-MSE scorer.

   What do the rows and columns of the returned arrays represent?

2. Average each returned score array across folds and negate the scores. Check that the resulting training and validation MSEs agree with Part 3 using `np.allclose()`.

[16]:
```
from sklearn.model_selection import validation_curve

# TODO: Call validation_curve() with param_name='poly__degree'.

# TODO: Average across folds, negate the scores, and compare with Part 3.
```

Solution.

`validation_curve()` handles the loop over degrees. Both returned arrays have shape `(15, 10)`: rows correspond to degrees and columns to folds.

[17]:
```
# Compute the validation curve by varying the polynomial degree poly__degree
train_scores, validate_scores = validation_curve(
    pipe,
    X,
    y,
    param_name='poly__degree',
    param_range=degrees,
    cv=cv_splitter,
    scoring='neg_mean_squared_error',
)
```

```
# Compute average scores across folds (folds are in columns, hence axis=1)
mse_train_curve = -train_scores.mean(axis=1)
mse_validate_curve = -validate_scores.mean(axis=1)
```

We can use `np.allclose()` to check that the mean training and validation MSEs agree with those computed in Part 3.

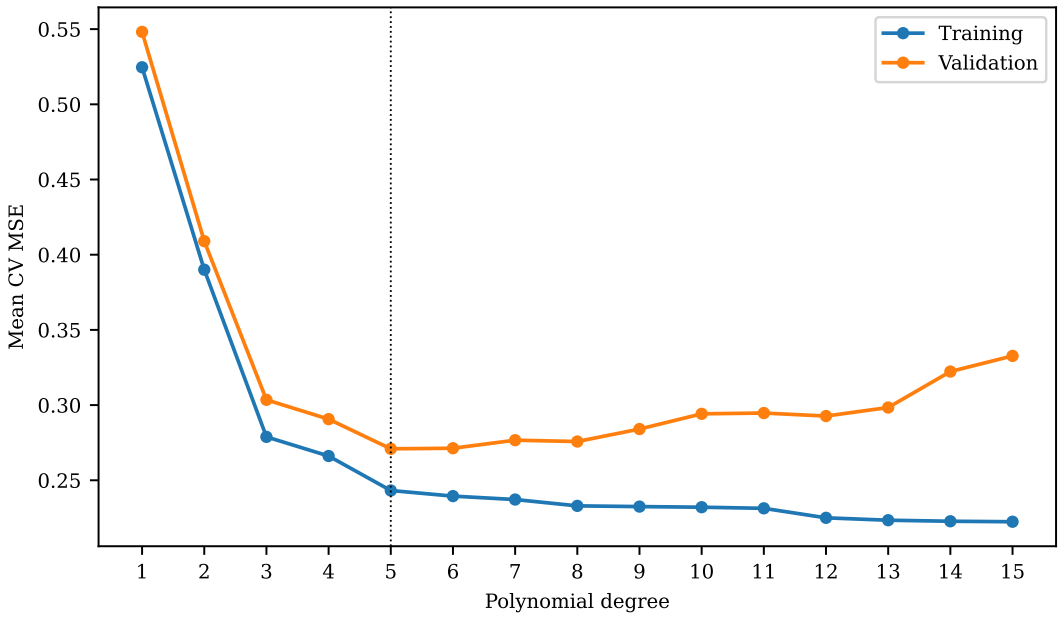
[18]:
```
print('Training MSEs are close:', np.allclose(mse_train_mean, mse_train_curve))
print('Validation MSEs are close:', np.allclose(mse_validate_mean, mse_validate_curve))
```

```
Training MSEs are close: True
Validation MSEs are close: True
```

For comparison, we can also draw the same validation curve using `ValidationCurveDisplay`. We construct the display from the scores already computed, so CV is not repeated. The argument `negate_score=True` displays positive MSEs.

[19]:
```
from sklearn.model_selection import ValidationCurveDisplay

display = ValidationCurveDisplay(
    param_name='poly__degree',
    param_range=degrees,
    train_scores=train_scores,
    test_scores=validate_scores,
    score_name='MSE',
)
fig, ax = plt.subplots(figsize=(6, 3.5), layout='constrained')
display.plot(
    ax=ax,
    negate_score=True,
    std_display_style=None,
    line_kw={'marker': 'o', 'markersize': 4},
)
ax.axvline(dmin, ls=':', lw=0.75, c='black')
ax.set(xlabel='Polynomial degree', ylabel='Mean CV MSE', xticks=degrees)
_ = ax.legend(['Training', 'Validation'])
```



Part 5 — Selecting and refitting automatically

Next, we want to select the polynomial degree and refit the model automatically using `GridSearchCV` instead of computing the validation curve manually.

1. Use `GridSearchCV` to select among the same degrees, using the same pipeline, folds, and scorer.

   Hint: Set `refit=True` to automatically refit the best model on the full training data.

2. Print the selected degree using the `best_params_` attribute and its mean validation MSE from the `best_score_` attribute of the `GridSearchCV` object. Remember to negate the score to obtain the MSE.

   Compare the printed values with the results from Part 3.

[20]:
```
from sklearn.model_selection import GridSearchCV

# TODO: Create grid_search with the same degree grid, splitter, and scorer.

# TODO: Fit the search on X and y, then print the selected degree and CV MSE.
```

Solution.

The parameter name `poly__degree` combines the pipeline step name and its parameter, separated by two underscores. With `refit=True`, the search fits the selected pipeline on all 100 training observations after comparing the CV scores.

[21]:
```
from sklearn.model_selection import GridSearchCV

# Define parameter grid
param_grid = {'poly__degree': degrees}

# Create grid_search with the same degree grid, splitter, and scorer.
grid_search = GridSearchCV(
    pipe,
    param_grid,  # parameters to search over
    cv=cv_splitter,  # cross-validation splitter
    scoring='neg_mean_squared_error',  # scoring metric
    refit=True,  # refit the best model on the whole dataset
    n_jobs=4,  # number of jobs to run in parallel
)

# Run the grid search on the data
grid_search.fit(X, y)
```

[21]:
```
GridSearchCV(cv=KFold(n_splits=10, random_state=1234, shuffle=True),
             estimator=Pipeline(steps=[('input_scale', StandardScaler()),
                                       ('poly',
                                        PolynomialFeatures(degree=np.int64(15),
                                                           include_bias=False)),
                                       ('scale', StandardScaler()),
                                       ('lr', LinearRegression())]),
             n_jobs=4,
             param_grid={'poly__degree': array([ 1,  2,  3,  4,  5,  6,  7,  8,
       9, 10, 11, 12, 13, 14, 15])},
             scoring='neg_mean_squared_error')
```

[22]:
```
# Extract the selected degree from the grid search results
selected_degree = grid_search.best_params_['poly__degree']

print(f'Selected degree: {selected_degree}')
print(f'Minimum CV MSE: {-grid_search.best_score_:.3f}')
```

```
Selected degree: 5
Minimum CV MSE: 0.271
```

The search again selects degree 5 with a minimum CV MSE of about 0.271. `best_estimator_` contains the refitted pipeline, ready to predict new observations. The copies fitted during CV used only nine of the ten folds each.

Part 6 — Comparing fitted models

You are now asked to visually compare the fitted models with degree 1, the CV-selected degree, and degree 15.

1. Fit the comparison models with degrees 1 and 15 on all training observations, and use `GridSearchCV`'s `best_estimator_` attribute for the selected best model.

   Hint: Instead of defining new pipelines, you can use `clone()` to create unfitted copies of the pipeline for degrees 1 and 15, and then adjust their `poly__degree` parameter using `set_params(poly__degree=...)`.

2. Use the supplied function below to plot a fitted model together with the sample and true function. Call it once for each model to create three separate plots.

3. Explain how the plots relate to the validation curve.

[23]:
```python
def plot_fitted_model(X, y, model, title):
    """
    Plot one fitted model with the sample and true function.

    Parameters
    ----------
    X : array
        Training feature matrix with one column
    y : array
        Training outcomes
    model : estimator
        Fitted regression model or pipeline
    title : str
        Plot title
    """
    ax = plot_sample(X, y)
    x_grid = np.linspace(0, 1, 301)
    y_hat = model.predict(x_grid[:, None])
    ax.plot(x_grid, y_hat, c='darkorange', lw=1.25, label='Fitted polynomial')
    ax.set_title(title)
    ax.legend()
```

[24]:
```python
from sklearn.base import clone

# Fit degree-1 copy of pipe.
model_d1 = clone(pipe).set_params(poly__degree=1).fit(X, y)

# TODO: Fit degree-15 copy of pipe.

# TODO: Save best model from grid_search.best_estimator_ as best_model

# TODO: Call plot_fitted_model() for each of the three models.
```

Solution.

We use the refitted `best_estimator_` for the selected degree and fit separate copies for the comparison models. All three plots use the same axis limits.

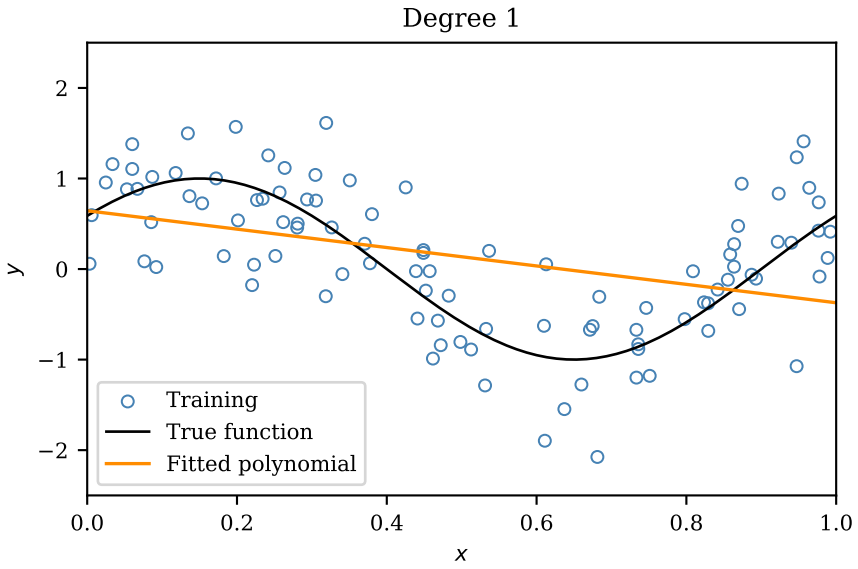
[25]:
```
from sklearn.base import clone

# Clone and fit the degree 1 model
model_d1 = clone(pipe).set_params(poly__degree=1).fit(X, y)

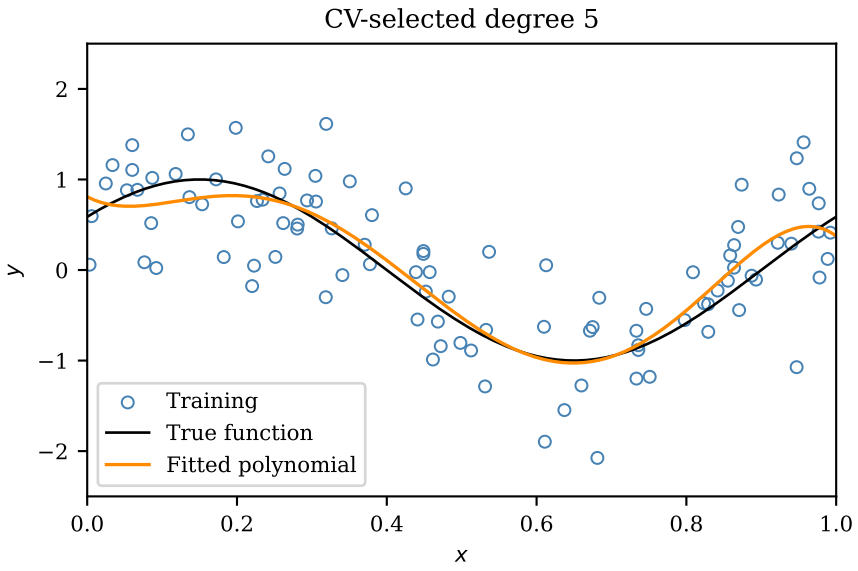
# Clone and fit the degree 15 model
model_d15 = clone(pipe).set_params(poly__degree=15).fit(X, y)

# Extract the best model from the grid search
best_model = grid_search.best_estimator_
```

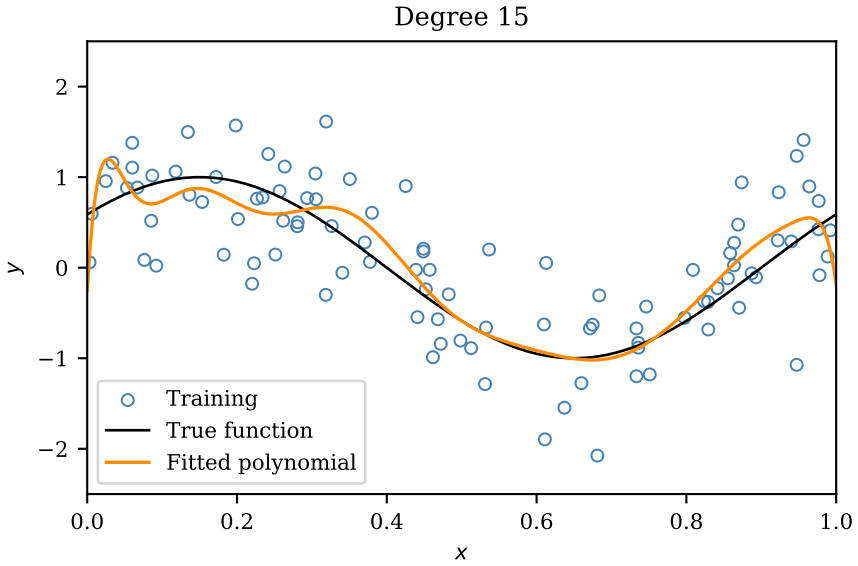
[26]:
```
plot_fitted_model(X, y, model_d1, 'Degree 1')
```



[27]:
```
title = f'CV-selected degree {selected_degree}'
plot_fitted_model(X, y, best_model, title)
```



[28]:
```
plot_fitted_model(X, y, model_d15, 'Degree 15')
```



The linear (degree 1) model misses the curved relationship. The selected model follows the true function more closely. Degree 15 fits the training observations more closely but has a higher CV error. Its fitted curve is somewhat more variable; the validation scores help us assess differences that may be hard to judge visually.

Part 7 — Independent evaluation

We used cross-validation to choose the polynomial degree and then fitted the selected model on all 100 training observations. We now evaluate predictions using a new test sample. These observations were not used to choose the degree or fit any of the models.

Because these data are simulated, we can generate a new test sample instead of setting aside some of the original observations.

1. Use `generate_sample()` to generate 10,000 independent test observations with `default_rng(5678)`, the same true function, and the same noise level.
2. Use `mean_squared_error()` to report training and test MSE for the three models fitted in Part 6.

Explain:

1. Which model has the lowest training MSE? Which has the lowest test MSE?
2. Compare degree 15 with the CV-selected degree. Does the lower training MSE translate into better predictions on the test sample? Explain the difference.

[29]:
```
import pandas as pd
from sklearn.metrics import mean_squared_error

# TODO: Generate the test sample X_test, y_test

# TODO: Compute and report each model's training and test MSE.
```

Solution.

We generate a new test sample and evaluate the three fitted models without fitting them again.

[30]:
```
# Initialize an RNG with a different seed to get new test sample
rng_test = default_rng(5678)

# Generate the test sample
X_test, y_test = generate_sample(10_000, sigma, rng_test)
```

[31]:
```
import pandas as pd
from sklearn.metrics import mean_squared_error

# Dictionary of models to evaluate
models = {1: model_d1, selected_degree: best_model, 15: model_d15}

rows = []

for degree, model in models.items():
    # Compute each model's training and test MSE
    train_mse = mean_squared_error(y, model.predict(X))
    test_mse = mean_squared_error(y_test, model.predict(X_test))

    rows.append(
        {'Model': f'Degree {degree}', 'Training MSE': train_mse, 'Test MSE': test_mse}
    )

# Build a DataFrame to report the training and test MSEs for each model
df_errors = pd.DataFrame(rows)
df_errors
```

[31]:

	Model	Training MSE	Test MSE
0	Degree 1	0.525792	0.614275
1	Degree 5	0.244521	0.273138
2	Degree 15	0.226415	0.294391

Degree 15 has the lowest training MSE (about 0.226), while the CV-selected degree 5 has the lowest test MSE (about 0.273).

Compared with degree 5, degree 15 reduces training MSE from about 0.245 to 0.226 but increases test MSE from about 0.273 to 0.294. Its additional flexibility lets it fit some of the random noise in the training sample. That noise does not help predict new observations, so its test MSE is higher.

Exercise 2: Categorical features

In this exercise, we practice fitting models that include categorical features in the regressor matrix $\mathbf{X}$ using the Ames housing dataset. There are several categorical variables in this data that come in two different kinds:

1. Nominal categorical variables capture qualitative categories. For numerical purposes, we often encode these as integer values, but any particular integer representation is arbitrary. It is a serious mistake to include such integer-valued categorical variables as *continuous* features in a model.

   In the Ames housing data, the `Neighborhood` column is one such nominal categorical variable (stored as a string containing the neighborhood name).

2. Ordinal categorical variables represent data that have a natural ordering, but the distance between any two values can be arbitrary.

   For example, in the Ames housing data the `OverallQuality` variable is an ordinal categorical variable on a scale of 1–10. While a value of 2 is clearly better than a value of 1, it is up to interpretation how much better.

Such variables can be included as categorical *dummy variables* in a model, not imposing any particular interpretation on the distance between two values. Alternatively, such variables can be included as *continuous* variables, which imposes how the distance is interpreted. For example, including an ordinal categorical variable as a linear term in a model imposes that moving from 1 to 2 has the same effect on the outcome variable as moving from 2 to 3.

Part 1 — Creating the estimation sample

We load the Ames data the same way as we did in the lectures:

[32]:
```
import pandas as pd

# Use local data/ folder
DATA_PATH = '../../data'

filename = f'{DATA_PATH}/ames_houses.csv'
df = pd.read_csv(filename, sep=',')
```

First, perform the following data preprocessing steps:

1. Drop all observations that have any missing values for `SalePrice`, `OverallQuality`, or `Neighborhood`.
2. Tabulate how many neighborhoods have fewer than 40 observations, and drop observations from these neighborhoods from the dataset.

Solution.

[33]:
```
# Keep only obs with non-missing data for relevant variables
variables = ['SalePrice', 'OverallQuality', 'Neighborhood']
df = df.dropna(subset=variables)
```

[34]:
```
# Compile list of neighborhoods with fewer than 40 observations
nobs = df['Neighborhood'].value_counts().sort_values()

# Select neighborhoods with fewer than minimum observations
MIN_OBS = 40
few_nobs = nobs.loc[nobs < MIN_OBS]

print(
    f'Number of neighborhoods with fewer than {MIN_OBS} observations: {len(few_nobs)}'
)
# Print these neighborhoods
few_nobs
```

```
Number of neighborhoods with fewer than 40 observations: 9
```

[34]:
```
Neighborhood
Landmrk      1
GrnHill      2
Greens       8
Blueste     10
NPkVill     23
Veenker     24
Blmngtn     28
BrDale      30
MeadowV     37
Name: count, dtype: int64
```

[35]:
```
# Keep only neighborhoods with minimum number of observations
nobs = df.groupby('Neighborhood')['Neighborhood'].transform('count')
df = df[nobs >= MIN_OBS]
```

Part 2 — Train-test split

Split the data into a training and a test sample, assigning 40% of observations to the test sample, and report the number of observations in each set.

Stratify your split by `Neighborhood` to preserve approximately the same neighborhood proportions in both samples. Use the `stratify` argument of `train_test_split()` and specify a random seed of 1234.

You can pass the whole `DataFrame` to `train_test_split()` instead of individual `X` and `y` values. The function then returns two DataFrames, one containing the training data and one containing the test data.

Solution.

[36]:
```
from sklearn.model_selection import train_test_split

# Fraction of sample to use for testing
test_size = 0.4

# Split sample into training and test samples using the whole DataFrame
df_train, df_test = train_test_split(
    df, test_size=test_size, stratify=df['Neighborhood'], random_state=1234
)

print(f'Number of observations in training sample: {len(df_train):5,d}')
print(f'Number of observations in test sample:     {len(df_test):5,d}')
```

```
Number of observations in training sample: 1,660
Number of observations in test sample:     1,107
```

Part 3 — Exploratory data analysis

Use only `df_train` for the following exploratory analysis so that test outcomes do not guide model development:

1. Compute the correlations of `SalePrice` with the other numerical variables. Which explanatory variable has the highest correlation?
2. Create a bar chart of `OverallQuality`, showing the number of observations for each quality level 1–10.

   Hint: Create a `Series` with the number of observations per quality level and plot it using `Series.plot.bar()`.
3. Create a box plot showing `SalePrice` by `OverallQuality`.

   Hint: You can use `DataFrame.plot.box()` for this.
4. Create a box plot showing `SalePrice` by `Neighborhood`.

Solution.

We can compute the pairwise correlations using the `corr()` method. This returns the whole (symmetric) correlation matrix, but since we are interested only in the correlations with `SalePrice`, we select only that row.

[37]:
```
# Compute training-sample correlations with the explanatory variables
df_train.corr(numeric_only=True).loc['SalePrice'].drop('SalePrice').sort_values(
    ascending=False
)
```

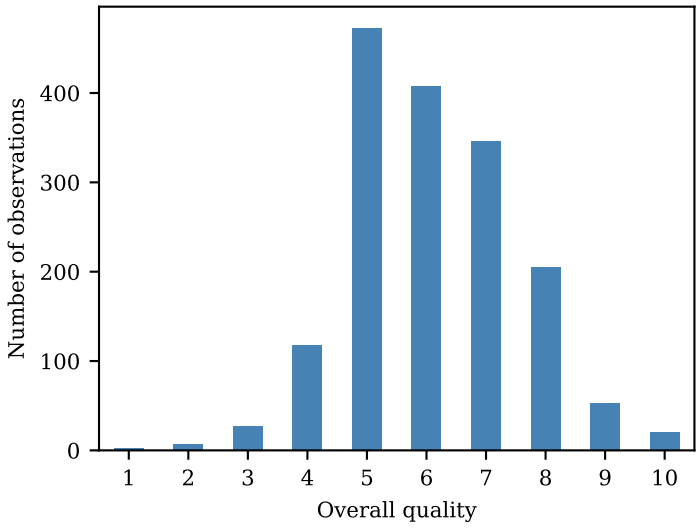
[37]:
```
OverallQuality          0.796639
LivingArea              0.715426
BasementArea            0.637367
YearBuilt               0.563902
Bathrooms               0.543826
YearRemodeled           0.532033
Fireplaces              0.471543
FinishedBasementArea    0.393978
LotArea                 0.241002
HasGarage               0.221328
Bedrooms                0.144113
MonthSold               0.047169
YearSold               -0.026633
OverallCondition       -0.121497
Name: SalePrice, dtype: float64
```

As the table shows, `OverallQuality` is the explanatory variable that has the highest correlation with `SalePrice`.

Next, we create a bar chart depicting the number of observations by overall quality. We can do this by grouping by `OverallQuality` and counting the number of elements with `size()` and then using the pandas plotting routines to create a bar chart.

[38]:
```
# Count training observations at each quality level
df_train.groupby('OverallQuality').size().reindex(range(1, 11), fill_value=0).plot.bar(
    rot=0, color='steelblue', xlabel='Overall quality', ylabel='Number of observations'
)
```

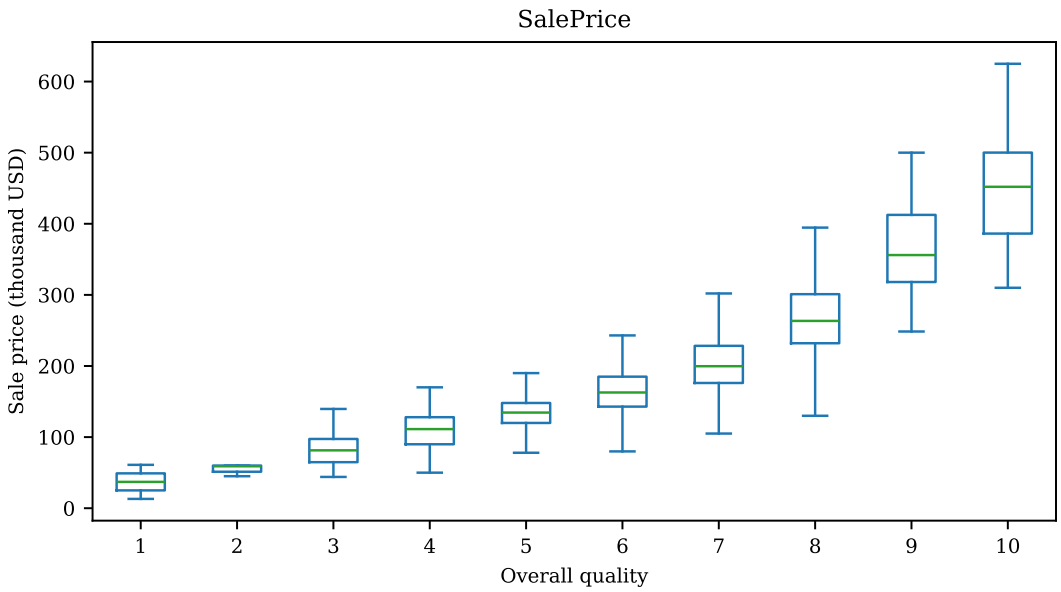
[38]:
```
<Axes: xlabel='Overall quality', ylabel='Number of observations'>
```



We use the `DataFrame.plot.box()` method to create a box plot of the sale price by overall quality.

[39]:
```
df_train.plot.box(
    by='OverallQuality',
    column='SalePrice',
    showfliers=False,
    figsize=(7, 3.5),
    ylabel='Sale price (thousand USD)',
    xlabel='Overall quality',
)
```

[39]:
```
SalePrice    Axes(0.125,0.11;0.775x0.77)
dtype: object
```

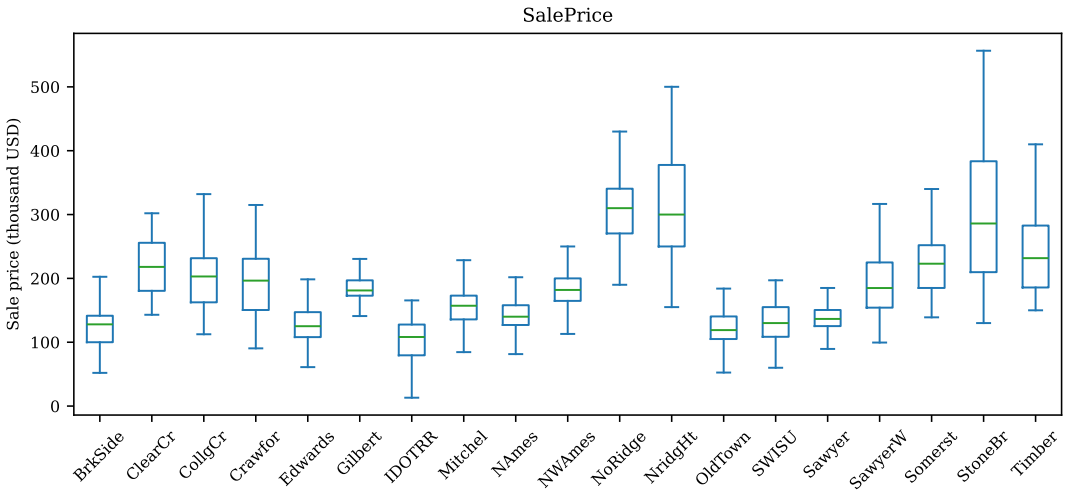


As you can see, there is a clear increasing relationship between sale price and overall quality.

The same approach can be used to create the box plot by neighborhood:

[40]:
```
df_train.plot.box(
    by='Neighborhood',
    column='SalePrice',
    showfliers=False,
    figsize=(9, 3.5),
    ylabel='Sale price (thousand USD)',
    xlabel='',
    rot=45,
)
```

[40]:
```
SalePrice    Axes(0.125,0.11;0.775x0.77)
dtype: object
```



Part 4 — Predicting price by neighborhood

Create an estimator pipeline to fit the model

$$SalePrice_{ij} = \mu_j + \epsilon_{ij}$$

where i indexes observations and j indexes neighborhoods. That is, the sale price for a house i in neighborhood j is predicted to be the average sale price μ_j in that neighborhood.

In order to estimate this model, you'll have to convert the categorical string variable `Neighborhood` to a set of 0/1 dummy variables. This can be achieved using the `OneHotEncoder` as a step in a pipeline.

Fit the model on the training data, and report the RMSE both on the training and the test sample.

This is a predetermined baseline. Use training data and CV scores to choose between the quality models in the remaining parts.

Solution.

We first extract the target and features from the training and test DataFrames. Note that scikit-learn expects a feature *matrix*, so we should extract the features as a list of columns even if there is only a single feature.

[41]:
```
features = ['Neighborhood']
target = 'SalePrice'

# Extract features and target variable
X_train, y_train = df_train[features], df_train[target]
X_test, y_test = df_test[features], df_test[target]
```

We can now build an estimation pipeline that uses `OneHotEncoder` as a data preprocessing step. This step converts the categorical variables into a matrix of zeros and ones, where each column corresponds to one neighborhood and contains 1 if the observation is located in that neighborhood, and 0 otherwise.

Note that we need to either drop the first category (`drop='first'`) if we want to fit the model with an intercept, or specify `fit_intercept=False`. Otherwise, the regressor matrix $\mathbf{X}$ will be collinear.

[42]:
```
from sklearn.linear_model import LinearRegression
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import OneHotEncoder

# Drop first dummy column, estimate model with intercept
```

```
pipe_dummies = make_pipeline(
    OneHotEncoder(drop='first', sparse_output=False),
    LinearRegression(fit_intercept=True),
)
```

Alternatively and equivalently, we can specify the estimation pipeline as follows:

[43]:
```
# Don't drop first dummy column, don't estimate model with additional intercept
pipe_dummies = make_pipeline(
    OneHotEncoder(drop=None, sparse_output=False), LinearRegression(fit_intercept=False)
)
```

[44]:
```
# Fit the model
pipe_dummies.fit(X_train, y_train)
```

[44]:
```
Pipeline(steps=[('onehotencoder', OneHotEncoder(sparse_output=False)),
                ('linearregression', LinearRegression(fit_intercept=False))])
```

With the fitted model, we can inspect the estimated coefficients:

[45]:
```
# Match each estimated neighborhood mean to its category label
neighborhoods = pipe_dummies.named_steps['onehotencoder'].categories_[0]
coefficients = pipe_dummies.named_steps['linearregression'].coef_
pd.Series(coefficients, index=neighborhoods, name='Mean price (thousand USD)')
```

[45]:
```
BrkSide    124.243077
ClearCr    216.938462
CollgCr    204.387500
Crawfor    201.872581
Edwards    134.881034
Gilbert    189.113131
IDOTRR     104.092857
Mitchel    162.120588
NAmes      145.115789
NWAmes     187.277215
NoRidge    332.820930
NridgHt    319.100000
OldTown    126.475524
SWISU      130.137931
Sawyer     136.491209
SawyerW    190.429333
Somerst    224.846789
StoneBr    308.777419
Timber     244.165116
Name: Mean price (thousand USD), dtype: float64
```

There are 19 estimated coefficients, one for each neighborhood in the data. The encoder's `categories_` attribute gives the labels in the order of the dummy columns. Each labeled estimate is the neighborhood's mean training sale price, measured in thousands of USD.

We use the fitted model to predict prices and compute RMSE.

[46]:
```
from sklearn.metrics import root_mean_squared_error

y_train_hat = pipe_dummies.predict(X_train)
y_test_hat = pipe_dummies.predict(X_test)

rmse_train = root_mean_squared_error(y_train, y_train_hat)
rmse_test = root_mean_squared_error(y_test, y_test_hat)

print(f'Training RMSE (thousand USD): {rmse_train:.5f}')
print(f'Test RMSE (thousand USD): {rmse_test:.5f}')
```

```
Training RMSE (thousand USD): 53.58493
```

```
Test RMSE (thousand USD): 52.82692
```

Part 5 — Polynomial in overall quality

Fit the model

$$SalePrice_i = \beta_0 + \beta_1 OverallQuality_i + \beta_2 OverallQuality_i^2 + \cdots + \beta_d OverallQuality_i^d + \epsilon_i$$

for polynomial degrees $d = 1, 2, \ldots, 9$. This treats quality as a numeric predictor and imposes a smooth relationship across its ordered levels. In the next part, we compare this with a separate mean for each quality category.

Use the template below and `cross_validate()` to compute mean training and validation RMSEs. Include `StandardScaler` both before and after `PolynomialFeatures(degree=d, include_bias=False)`, followed by `LinearRegression(fit_intercept=True)`. Keep the scalers inside the pipeline.

Use `scoring='neg_root_mean_squared_error'` and `return_train_score=True`. The arrays under `train_score` and `test_score` contain negative RMSEs for each fold; `test_score` refers to the CV validation folds.

Plot the mean training and validation RMSEs and report the degree with the smallest mean validation RMSE. Reuse `cv_quality` for the dummy-model comparison. The training observations are already in random order after the train-test split, so we keep that order when assigning the five folds.

[47]:
```python
import numpy as np
from sklearn.model_selection import KFold

features = ['OverallQuality']
target = 'SalePrice'
X_train, y_train = df_train[features], df_train[target]
X_test, y_test = df_test[features], df_test[target]

degrees = np.arange(1, 10)
cv_quality = KFold(n_splits=5)
rmse_train_mean = []
rmse_validate_mean = []

for d in degrees:
    # TODO: Create the scaling + polynomial features + regression pipeline.

    # TODO: Compute training and validation scores using cross_validate().
    # scores = cross_validate(
    #     estimator=...,
    #     X=X_train, y=y_train,
    #     scoring=...,
    #     cv=cv_quality,
    #     return_train_score=True,
    #     n_jobs=4
    # )

    # TODO: Negate and average the scores across folds.
    # rmse_train_mean.append(np.mean(-scores['train_score']))
    # rmse_validate_mean.append(np.mean(-scores['test_score']))

    pass
```

Solution.

The template selects `OverallQuality` as a one-column feature matrix. We scale the input before creating powers and scale the expanded columns before regression, using the same pipeline structure as in Exercise 1.

We create a pipeline for each degree and run CV with the shared splitter. The chosen scorer returns negative RMSE, so we negate and average the scores across folds. All RMSEs are measured in thousands of USD.

[48]:
```
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.metrics import root_mean_squared_error
from sklearn.model_selection import cross_validate
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import PolynomialFeatures, StandardScaler

# Polynomial degrees to consider
degrees = np.arange(1, 10)

rmse_train_mean = []
rmse_validate_mean = []

for d in degrees:
    # Create polynomial features + linear regression pipeline
    pipe_lr = make_pipeline(
        StandardScaler(),
        PolynomialFeatures(degree=d, include_bias=False),
        StandardScaler(),
        LinearRegression(fit_intercept=True),
    )

    # Run cross-validation. Function returns a dictionary with scores
    scores = cross_validate(
        estimator=pipe_lr,  # Estimator to use
        X=X_train,
        y=y_train,  # Data to fit
        scoring='neg_root_mean_squared_error',  # Scoring metric
        cv=cv_quality,  # Shared five-fold splitter
        return_train_score=True,  # Also return training scores
        n_jobs=4,  # Fit up to four folds in parallel
    )

    # Compute average RMSE over all CV splits
    rmse_train_mean.append(np.mean(-scores['train_score']))
    rmse_validate_mean.append(np.mean(-scores['test_score']))

# Convert to NumPy arrays
rmse_train_mean = np.array(rmse_train_mean)
rmse_validate_mean = np.array(rmse_validate_mean)
```

Using this data, we can determine the optimal polynomial degree as the one that minimizes the mean validation RMSE across folds.

[49]:
```
# Find degree that minimizes RMSE
imin = np.argmin(rmse_validate_mean)
dmin = degrees[imin]

print(f'Polynomial degree with min. RMSE: {dmin}')
```

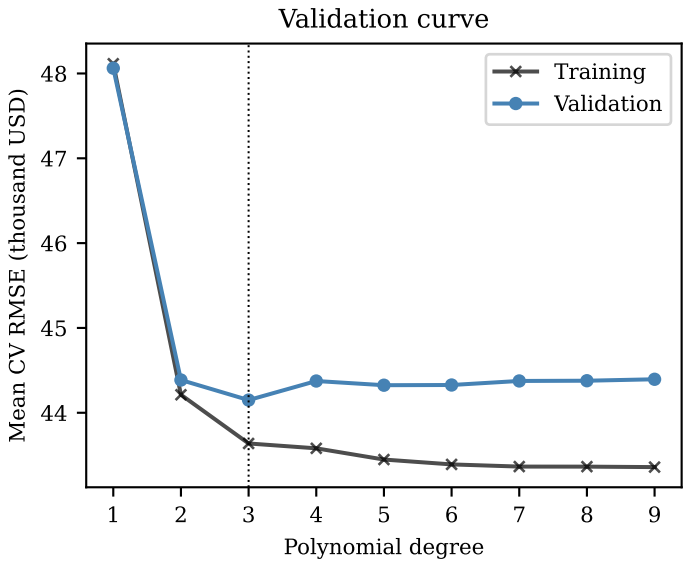
```
Polynomial degree with min. RMSE: 3
```

Finally, the following code creates the validation curve.

[50]:
```
import matplotlib.pyplot as plt

# Plot validation curve
plt.plot(
    degrees, rmse_train_mean, c='black', marker='x', ms=4, alpha=0.7, label='Training'
)
plt.plot(
    degrees, rmse_validate_mean, c='steelblue', marker='o', ms=4, label='Validation'
)
plt.xlabel('Polynomial degree')
plt.ylabel('Mean CV RMSE (thousand USD)')
plt.xticks(degrees)
plt.axvline(dmin, ls=':', lw=0.75, c='black')
plt.title('Validation curve')
plt.legend()
```

[50]:
```
<matplotlib.legend.Legend at 0x7f15459746e0>
```



Part 6 — Quality dummies

Compare the polynomial models with a separate mean for each quality category:

$$SalePrice_{ij} = \mu_j + \epsilon_{ij},$$

where j indexes the quality levels 1–10. A house with quality j is predicted to have the average training sale price of houses with that quality.

Create a pipeline with `OneHotEncoder` and `LinearRegression`, then evaluate it using the same observations, `cv_quality` splitter, and RMSE scorer as in Part 5. The dummy model has no polynomial degree to tune, but CV lets us compare its predictive performance with the polynomial models.

Recreate the validation curve and add horizontal lines for the dummy model's mean training and validation RMSEs. Compare its scores with those of degree 9. Why might these agree when all ten quality levels occur in every training fold?

Choose between the best polynomial model and the dummy model using their mean validation RMSEs, before inspecting their test predictions.

[51]:
```
# TODO: Create pipeline with dummy variables + estimation

# TODO: Run cross-validation
# scores = cross_validate(
#     estimator=...,
#     X=X_train, y=y_train,
#     scoring=...,
#     cv=cv_quality,
#     return_train_score=True,
#     n_jobs=4
# )

# TODO: Compute average RMSE over all CV splits
# rmse_train_dummies_mean = np.mean(-scores['train_score'])
# rmse_validate_dummies_mean = np.mean(-scores['test_score'])
```

Solution.

[52]:
```
from sklearn.preprocessing import OneHotEncoder

# Create pipeline with dummy variables + linear regression
pipe_dummies = make_pipeline(
    OneHotEncoder(drop='first'), LinearRegression(fit_intercept=True)
)

# Run cross-validation
scores = cross_validate(
    estimator=pipe_dummies,
    X=X_train,
    y=y_train,
    scoring='neg_root_mean_squared_error',
    cv=cv_quality,
    return_train_score=True,
    n_jobs=4,
)

# Compute average RMSE over all CV splits
rmse_train_dummies_mean = np.mean(-scores['train_score'])
rmse_validate_dummies_mean = np.mean(-scores['test_score'])

print(f'Average RMSE on train sample: {rmse_train_dummies_mean:.5f}')
print(f'Average RMSE on validation sample: {rmse_validate_dummies_mean:.5f}')

print(
    'Degree-9 training RMSE agrees:',
    np.isclose(rmse_train_mean[-1], rmse_train_dummies_mean),
)
print(
    'Degree-9 validation RMSE agrees:',
    np.isclose(rmse_validate_mean[-1], rmse_validate_dummies_mean),
)

if rmse_validate_mean[imin] < rmse_validate_dummies_mean:
    preferred_model = f'Polynomial degree {dmin}'
else:
    preferred_model = 'Quality dummies'
print(f'Preferred model using CV: {preferred_model}')
```

```
Average RMSE on train sample: 43.35949
Average RMSE on validation sample: 44.39483
Degree-9 training RMSE agrees: True
Degree-9 validation RMSE agrees: True
```

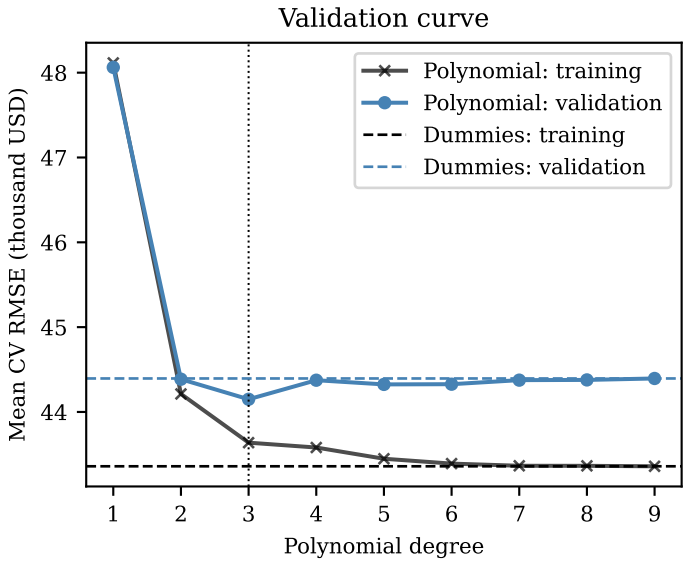
```
Preferred model using CV: Polynomial degree 3
```

We use the same plotting code as above, and add the horizontal lines for the dummy variable model on top.

[53]:
```
# Plot validation curve
plt.plot(
    degrees,
    rmse_train_mean,
    c='black',
    marker='x',
    ms=4,
    alpha=0.7,
    label='Polynomial: training',
)
plt.plot(
    degrees,
    rmse_validate_mean,
    c='steelblue',
    marker='o',
    ms=4,
    label='Polynomial: validation',
)
plt.xlabel('Polynomial degree')
plt.ylabel('Mean CV RMSE (thousand USD)')
plt.xticks(degrees)
plt.axvline(dmin, ls=':', lw=0.75, c='black')

# Add horizontal lines for RMSE from dummy variable regressions
plt.axhline(
    rmse_train_dummies_mean, ls='--', lw=1.0, c='black', label='Dummies: training'
)
plt.axhline(
    rmse_validate_dummies_mean,
    ls='--',
    lw=1.0,
    c='steelblue',
    label='Dummies: validation',
)
plt.title('Validation curve')
plt.legend()
```

[53]:
```
<matplotlib.legend.Legend at 0x7f15459e5a90>
```



The dummy model estimates a separate mean for each quality level, so it can fit the training data at least as closely as the polynomial models. With ten distinct quality levels, a degree-9 polynomial with an intercept can reproduce any set of ten fitted category means. All ten levels occur in each training fold here, so the degree-9 and dummy models have the same scores up to numerical precision: training RMSE is about 43.35949 and validation RMSE is about 44.39483.

Greater flexibility does not guarantee better predictions. Separate category means can be estimated imprecisely when a quality level has few observations. A low-degree polynomial restricts how predicted prices vary across levels.

CV favors degree 3, with mean validation RMSE about 44.14963, over the dummy model's 44.39483. We choose the cubic before evaluating the two models on the test sample.

Part 7 — Comparing test predictions

Refit the selected polynomial and the dummy model on the whole training sample and compute their test RMSEs. Report the results for both predetermined models.

Does the test ranking agree with the CV ranking from Part 6? How large is the difference in test RMSE? Discuss what this comparison tells us without revising the model choice using the test results.

Solution.

[54]:
```
# Re-estimate polynomial model on the whole training sample
pipe_poly_best = make_pipeline(
    StandardScaler(),
    PolynomialFeatures(degree=dmin, include_bias=False),
    StandardScaler(),
    LinearRegression(fit_intercept=True),
).fit(X_train, y_train)

# Re-estimate dummies model on the whole training sample
pipe_dummies.fit(X_train, y_train)

# Predict response variable on test sample
y_test_poly_hat = pipe_poly_best.predict(X_test)
y_test_dummies_hat = pipe_dummies.predict(X_test)
```

```python
# Compute RMSE on test sample
rmse_test_poly = root_mean_squared_error(y_test, y_test_poly_hat)
rmse_test_dummies = root_mean_squared_error(y_test, y_test_dummies_hat)

print(f'RMSE on test sample (polynomial): {rmse_test_poly:.5f}')
print(f'RMSE on test sample (dummies): {rmse_test_dummies:.5f}')
```

```
RMSE on test sample (polynomial): 42.93533
RMSE on test sample (dummies): 42.82456
```

Test RMSE is about 42.93533 for the cubic and 42.82456 for the dummy model, measured in thousands of USD. The dummy model has the smaller test RMSE on this split, reversing the CV ranking. The difference is only about 0.111 thousand USD, or $111, roughly 0.26% of the cubic's test RMSE.

The two samples give slightly different rankings. This small test difference does not establish that the dummy representation is generally better. Our model choice used CV; the test results report its performance after selection.

Exercise 3: Predicting loan interest rates with PCR

In this exercise, we build a model to predict a loan's interest rate from loan and borrower characteristics. We use principal component regression (PCR) to combine related predictors, and cross-validation to decide whether polynomial features and dimension reduction improve predictions.

We use data on 10,000 funded loans from OpenIntro's Lending Club dataset, stored in the file `lending_club.csv`. The loans were issued in the first quarter of 2018.

Our target is `interest_rate`, the annual interest rate in percent. We use the following eleven predictors. Counts and loan term enter as numeric variables. Homeownership and income verification enter as indicators.

Predictor	Description
`loan_amount`	Loan principal in US dollars
`term`	Loan term, either 36 or 60 months
`annual_income`	Applicant's annual income in US dollars
`debt_to_income`	Monthly debt payments as a percentage of monthly income
`homeowner`	1 for mortgage or outright ownership; 0 for renting
`income_verified`	1 if the source or amount of income was verified; 0 otherwise
`credit_history_years`	Issue year minus earliest credit-line year
`inquiries_last_12m`	Number of credit inquiries in the preceding 12 months
`open_credit_lines`	Number of open credit lines
`total_credit_limit`	Existing credit limit excluding mortgages, in US dollars
`credit_utilization`	Existing credit balance / limit, excluding mortgages

Credit utilization is a fraction: 0.5 means 50%. Both the balance and the limit exclude mortgages. These variables describe existing credit accounts across lenders, including accounts outside Lending Club. We interpret them as the applicant's credit position at application, before the loan in that row is issued.

Setup

We load the data and define the predictor list below. The setup keeps individual applications so that income and debt-to-income describe one applicant's finances. It removes observations missing the target or a selected predictor. The resulting DataFrame, `df_data`, contains 8,503 loans.

All model development will use the training sample; the test sample is reserved for Part 5.

[55]:
```python
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

# Path to CSV in data/ folder
DATA_PATH = '../../data/lending-club'
df_raw = pd.read_csv(f'{DATA_PATH}/lending_club.csv')

# Select features and target variable for regression
features = [
    'loan_amount',
    'term',
    'annual_income',
    'debt_to_income',
    'homeowner',
    'income_verified',
    'credit_history_years',
    'inquiries_last_12m',
    'open_credit_lines',
    'total_credit_limit',
    'credit_utilization',
]
target = 'interest_rate'

# Keep individual applications (drop joint applications)
df_individual = df_raw.query("application_type == 'individual'")
# Keep only observations with non-missing values for the selected features and target
df_data = df_individual.dropna(subset=[*features, target])

print(f'Loans in the teaching sample: {len(df_raw):,}')
print(f'Usable observations: {len(df_data):,}')
```

```
Loans in the teaching sample: 10,000
Usable observations: 8,503
```

Part 1 — Splitting and inspecting the sample

We first set aside a test sample to evaluate predictions on unseen loans. We then inspect the training predictors' scales and correlations to see why standardization and PCA may be useful.

1. Use the prepared `df_data` sample and `train_test_split()` to reserve 30% for testing, with `random_state=1234`. Name the resulting objects `X_train`, `X_test`, `y_train`, and `y_test`. Report the number of training and test observations.
2. Using only the training sample, report each predictor's mean, standard deviation, minimum, and maximum.

   Why would PCA on the unscaled predictors be sensitive to their units?
3. Create the correlation heatmap by calling `plot_correlations(X_train)` using the provided function below. It computes the correlation matrix and reads the predictor names from `X_train` automatically.

[56]:
```python
def plot_correlations(X):
    """
    Plot an annotated heatmap of predictor correlations.

    Parameters
    ----------
    X : dataframe
        Training predictors with named columns
    """
```

```
    df_corr = X.corr()

    fig, ax = plt.subplots(figsize=(6, 6), layout='constrained')
    heatmap = ax.imshow(df_corr, cmap='RdBu_r', vmin=-1, vmax=1)

    # Label the rows and columns with the predictor names
    positions = range(len(df_corr.columns))
    ax.set_xticks(positions, labels=df_corr.columns, rotation=45, ha='right')
    ax.set_yticks(positions, labels=df_corr.columns)

    # Display each correlation inside its cell
    for i in positions:
        for j in positions:
            correlation = df_corr.iloc[i, j]
            text_color = 'white' if abs(correlation) > 0.5 else 'black'
            ax.text(
                j,
                i,
                f'{correlation:.2f}',
                ha='center',
                va='center',
                c=text_color,
                fontsize=8,
            )

    fig.colorbar(heatmap, ax=ax, label='Correlation', shrink=0.5)
    ax.set_title('Training predictor correlations')
    plt.show()
```

[57]:
```
from sklearn.model_selection import train_test_split

# TODO: Split df_data into training and test sets and report their sizes.

# TODO: Inspect training predictor descriptive statistics

# TODO: Create correlation heatmap using plot_correlations(X_train).
```

Solution.

The setup provides 8,503 complete individual applications. We allocate 5,952 to training and 2,551 to testing.

[58]:
```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    df_data[features], df_data[target], test_size=0.3, random_state=1234
)

print(f'Training observations: {len(y_train):,}')
print(f'Test observations: {len(y_test):,}')
```

```
Training observations: 5,952
Test observations: 2,551
```

[59]:
```
df_summary = X_train.agg(['mean', 'std', 'min', 'max']).T
df_summary.round(2)
```

[59]:

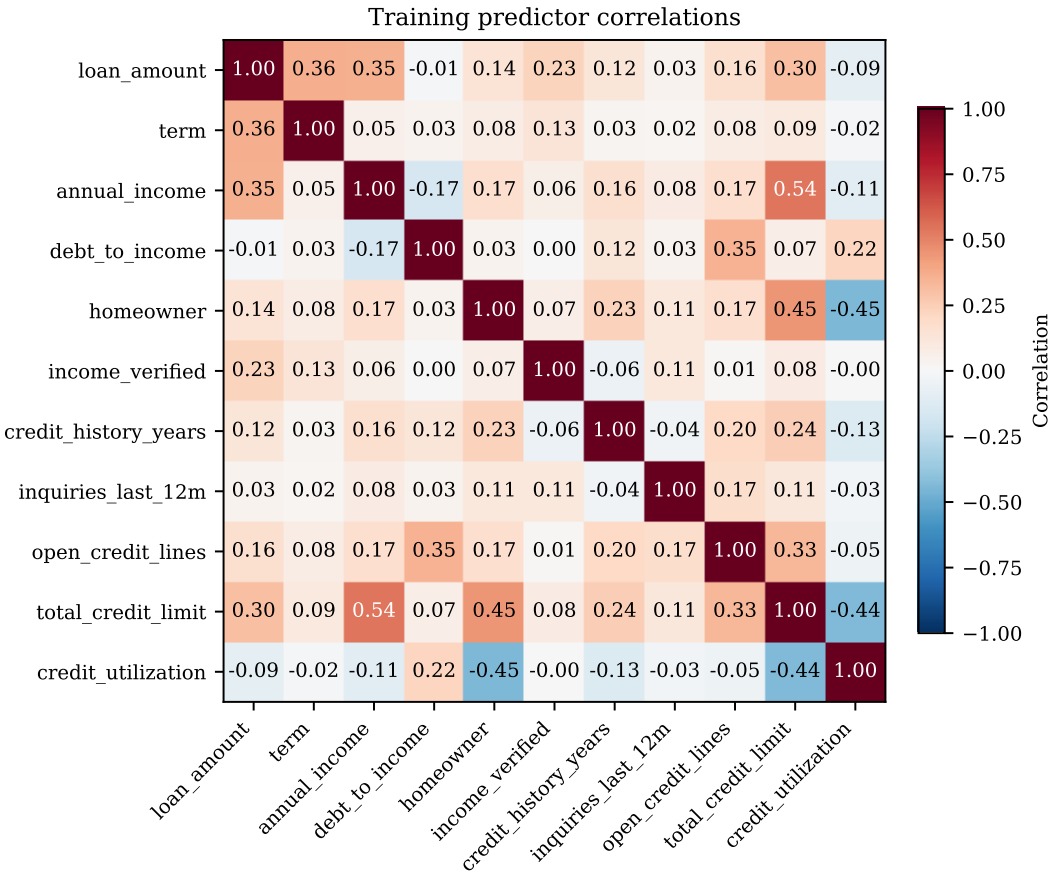
	mean	std	min	max
loan_amount	15675.98	10077.41	1000.0	40000.00
term	42.72	10.78	36.0	60.00
annual_income	83199.14	71465.19	5235.0	2300000.00
debt_to_income	17.15	8.79	0.0	39.99

homeowner	0.59	0.49	0.0	1.00
income_verified	0.62	0.48	0.0	1.00
credit_history_years	16.51	7.77	3.0	55.00
inquiries_last_12m	1.93	2.38	0.0	29.00
open_credit_lines	11.36	5.91	1.0	51.00
total_credit_limit	175846.21	190947.22	300.0	3386034.00
credit_utilization	0.41	0.27	0.0	1.84

Dollar amounts vary on a much larger numerical scale than counts or fractions. Without standardization, these units would strongly influence PCA's directions of greatest variation. Changing USD to thousands of USD could then change the components.

For each indicator, the training mean is the share of observations coded as 1. For example, the mean of `homeowner` is the share of applicants who own their home outright or with a mortgage.

[60]:
```
plot_correlations(X_train)
```



The heatmap shows positive correlations in red and negative correlations in blue; white indicates a correlation close to zero. Each diagonal entry is 1 because a predictor is perfectly correlated with itself.

The figure shows that there is moderate correlation between some of the predictors. The highest correlation is between annual income and total credit limit, with a training correlation of about 0.538. This correlation between predictor variables suggests that PCA may be useful to reduce the dimensionality of the data and capture the underlying structure.

Part 2 — Building the PCR pipeline

We next combine preprocessing, PCA, and regression in a single pipeline.

Build a `Pipeline` called `pipe_pcr` with these named steps:

1. `input_scale`: `StandardScaler()`.
2. `poly`: `PolynomialFeatures()` with `degree=2` and `include_bias=False`.
3. `scale`: another `StandardScaler()`.
4. `pca`: `PCA` with `n_components=0.9` and `svd_solver='full'`.
5. `lr`: `LinearRegression()` with `fit_intercept=True`.

Fit the pipeline on the training sample. Report the number of expanded features, the number of retained components, and their total explained predictor variance.

Hint: Use `n_features_in_`, `n_components_`, and `explained_variance_ratio_` from the fitted PCA step.

[61]:
```
from sklearn.decomposition import PCA
from sklearn.linear_model import LinearRegression
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import PolynomialFeatures, StandardScaler

# TODO: Build and fit pipe_pcr.

# TODO: Inspect the fitted PCA step and report its dimension and explained variance.
```

Solution.

The first scaler standardizes inputs before taking powers; the second standardizes the expanded features before PCA. Keeping both scalers and PCA inside the pipeline ensures that each CV fit learns these transformations from its training folds.

[62]:
```
from sklearn.decomposition import PCA
from sklearn.linear_model import LinearRegression
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import PolynomialFeatures, StandardScaler

# Pipeline for PCR
pipe_pcr = Pipeline(
    steps=[
        ('input_scale', StandardScaler()),
        ('poly', PolynomialFeatures(degree=2, include_bias=False)),
        ('scale', StandardScaler()),
        ('pca', PCA(n_components=0.9, svd_solver='full')),
        ('lr', LinearRegression(fit_intercept=True)),
    ]
)

# Fit the pipeline to the training data
pipe_pcr.fit(X_train, y_train)
```

[62]:
```
Pipeline(steps=[('input_scale', StandardScaler()),
                ('poly', PolynomialFeatures(include_bias=False)),
                ('scale', StandardScaler()),
                ('pca', PCA(n_components=0.9, svd_solver='full')),
                ('lr', LinearRegression())])
```

[63]:
```
# Store a reference to the fitted PCA step for inspection
initial_pca = pipe_pcr.named_steps['pca']

# Report the number of features and retained components
print(f'Expanded features: {initial_pca.n_features_in_}')
print(f'Retained components: {initial_pca.n_components_}')
```

```
# Report the proportion of variance retained by the PCA step
var_explained = initial_pca.explained_variance_ratio_.sum()
print(f'Predictor variance retained: {var_explained:.3f}')
```

```
Expanded features: 77
Retained components: 46
Predictor variance retained: 0.904
```

The expansion has 77 columns: eleven predictors, eleven squares, and 55 pairwise interactions. PCA retains 46 components, explaining about 90.4% of predictor variance. This percentage concerns the predictors, rather than interest rates.

Part 3 — Selecting degree and retention

We now use cross-validation to choose the polynomial degree and the amount of predictor variance retained by PCA.

Use `GridSearchCV` to select the optimal polynomial degree and PCA variance threshold jointly.

- Create a parameter grid with the following values:
  - Candidate degrees: `[1, 2, 3]`.
  - Candidate variance thresholds: `[0.7, 0.8, 0.9, 0.95, 0.99, 0.999, 0.9999]`.
- Use R^2 as the scoring metric, `scoring='r2'`.
- Define a CV splitter using `KFold` with 5 shuffled folds and a fixed seed of 1234.
- Create the `GridSearchCV` object with the argument `refit=True` so that the best model is refitted on the full training sample after the search.

Save the best model as `best_pcr` and call the provided `tabulate_cv_results(pcr_cv)` function to present the CV R^2 scores in a table with degrees as rows and variance thresholds as columns.

Use the table to discuss whether greater polynomial flexibility or more retained variance always improves prediction.

[64]:
```
def tabulate_cv_results(pcr_cv):
    """
    Tabulate the CV results from a GridSearchCV object for PCR with polynomial features.

    Parameters
    ----------
    pcr_cv : GridSearchCV
        Fitted GridSearchCV object

    Returns
    -------
    results : DataFrame
        Mean CV R² by polynomial degree and PCA variance threshold
    """
    keys = ['param_poly__degree', 'param_pca__n_components', 'mean_test_score']
    results = pd.DataFrame(pcr_cv.cv_results_)[keys]
    results = results.rename(
        columns={
            'mean_test_score': 'CV R²',
            'param_poly__degree': 'Poly degree',
            'param_pca__n_components': 'PCA variance',
        }
    )
```

```
    results = results.pivot(
        index='Poly degree', columns='PCA variance', values='CV R²'
    ).round(3)

    return results
```

[65]:
```
from sklearn.model_selection import GridSearchCV, KFold

# Candidate polynomial degrees and PCA variance thresholds
degrees = [1, 2, 3]
thresholds = [0.7, 0.8, 0.9, 0.95, 0.99, 0.999, 0.9999]
# Scoring metric for GridSearchCV
scoring = 'r2'

# TODO: Define cv_splitter, the parameter grid, and pcr_cv.

# TODO: Fit the search and report the selected parameters and mean CV R².

# TODO: Save best_estimator_ as best_pcr.
```

Solution.

We compare 21 parameter combinations on the same five folds and select the highest mean validation R^2.

[66]:
```
from sklearn.model_selection import GridSearchCV, KFold

# Candidate polynomial degrees and PCA variance thresholds
degrees = [1, 2, 3]
thresholds = [0.7, 0.8, 0.9, 0.95, 0.99, 0.999, 0.9999]

# Scoring metric for GridSearchCV
scoring = 'r2'

# Define a cross-validation splitter with 5 shuffled folds and a fixed random state
cv_splitter = KFold(n_splits=5, shuffle=True, random_state=1234)
```

We fit the grid search on the training sample:

[67]:
```
# Create dictionary of parameters to search over for GridSearchCV
param_grid = {'poly__degree': degrees, 'pca__n_components': thresholds}

pcr_cv = GridSearchCV(
    pipe_pcr, param_grid, cv=cv_splitter, scoring=scoring, refit=True, n_jobs=4
)
pcr_cv.fit(X_train, y_train)
```

[67]:
```
GridSearchCV(cv=KFold(n_splits=5, random_state=1234, shuffle=True),
             estimator=Pipeline(steps=[('input_scale', StandardScaler()),
                                       ('poly',
                                        PolynomialFeatures(include_bias=False)),
                                       ('scale', StandardScaler()),
                                       ('pca',
                                        PCA(n_components=0.9,
                                            svd_solver='full')),
                                       ('lr', LinearRegression())]),
             n_jobs=4,
             param_grid={'pca__n_components': [0.7, 0.8, 0.9, 0.95, 0.99, 0.999,
                                               0.9999],
                         'poly__degree': [1, 2, 3]},
             scoring='r2')
```

We extract the refitted best model and report its parameters and CV score:

[68]:
```
# Store the best estimator from the grid search
best_pcr = pcr_cv.best_estimator_

# Print the selected parameters and mean CV R²
print(pcr_cv.best_params_)
print(f'PCR CV selection R²: {pcr_cv.best_score_:.3f}')
```

```
{'pca__n_components': 0.999, 'poly__degree': 2}
PCR CV selection R²: 0.305
```

The search selects degree 2 and a variance threshold of 0.999, with mean CV R^2 about 0.305. We tabulate all combinations below.

[69]:
```
tabulate_cv_results(pcr_cv)
```

[69]:

PCA variance	0.7000	0.8000	0.9000	0.9500	0.9900	0.9990	0.9999
Poly degree							
1	0.239	0.242	0.293	0.294	0.294	0.294	0.294
2	0.274	0.281	0.273	0.283	0.301	0.305	0.302
3	0.243	0.274	0.275	0.266	0.272	0.235	0.191

More flexibility or retained variance does not always improve prediction. At degree 2, raising the variance threshold from 0.999 to 0.9999 lowers CV R^2 from about 0.305 to 0.302. Degree 3 reaches only about 0.275.

Part 4 — Comparing prediction methods

We now compare PCR with polynomial regression without dimension reduction. Does PCR improve predictions relative to using the polynomial features directly?

1. Build a new `Pipeline` called `pipe_poly` with these named steps:
    1. `input_scale`: `StandardScaler()`,
    2. `poly`: `PolynomialFeatures()` with `degree=2` and `include_bias=False`,
    3. `scale`: another `StandardScaler()`, and
    4. `lr`: `LinearRegression()` with `fit_intercept=True`.
2. Run a new `GridSearchCV` search called `poly_cv` over degrees 1, 2, and 3, using the same training observations, `cv_splitter`, and `scoring`.

    Save the refitted best model as `best_poly`.
3. Use the provided `compare_cv_results(pcr_cv, poly_cv)` function to compare the highest PCR CV R^2 across variance thresholds with the CV R^2 without PCA for each degree. Does discarding components help more at some degrees than at others?

Based on the validation results, which method would you choose for predicting interest rates?

[70]:
```
def compare_cv_results(pcr_cv, poly_cv):
    """
    Compare validation scores with and without PCA at each polynomial degree.

    Parameters
    ----------
    pcr_cv : search
        Fitted PCR grid search
    poly_cv : search
        Fitted polynomial regression grid search
```

```
    Returns
    -------
    dataframe
        Best PCR and polynomial CV scores at each degree
    """
    df_pcr = pd.DataFrame(pcr_cv.cv_results_)
    pcr_scores = df_pcr.groupby('param_poly__degree')['mean_test_score'].max()

    df_poly = pd.DataFrame(poly_cv.cv_results_)
    poly_scores = df_poly.set_index('param_poly__degree')['mean_test_score']

    df_compare = pd.DataFrame({'PCR': pcr_scores, 'Without PCA': poly_scores})
    df_compare.index.name = 'Degree'
    return df_compare.round(3)
```

[71]:
```
# TODO: Build pipe_poly

# TODO: Run GridSearchCV over degrees 1, 2, and 3.

# TODO: Save the best estimator as best_poly

# TODO: Call compare_cv_results(pcr_cv, poly_cv) and interpret the table.
```

Solution.

We use the same preprocessing, folds, and scorer, omitting the PCA step.

[72]:
```
# Define pipeline for polynomial regression without PCA
pipe_poly = Pipeline(
    steps=[
        ('input_scale', StandardScaler()),
        ('poly', PolynomialFeatures(degree=2, include_bias=False)),
        ('scale', StandardScaler()),
        ('lr', LinearRegression(fit_intercept=True)),
    ]
)

# Set up and fit GridSearchCV to search over polynomial degrees
poly_cv = GridSearchCV(
    pipe_poly, {'poly__degree': degrees}, cv=cv_splitter, scoring=scoring, refit=True
)
poly_cv.fit(X_train, y_train)

# Store the best estimator from the grid search
best_poly = poly_cv.best_estimator_
```

We can then use the provided function to compare the best CV R^2 for each degree with and without PCA:

[73]:
```
# Tabulate the best CV scores for each polynomial degree
compare_cv_results(pcr_cv, poly_cv)
```

[73]:

Degree	PCR	Without PCA
1	0.294	0.294
2	0.305	0.302
3	0.275	-0.132

Both searches select degree 2. PCR has a slightly higher CV R^2: about 0.305 compared with 0.302 for polynomial regression. Based on this validation comparison, we would choose PCR for predicting interest rates.

At degree 3, PCA improves CV R^2 from about -0.132 to 0.275, highlighting the benefit of discarding components when the polynomial expansion is large and multicollinear.

Part 5 — Evaluating unseen loans

We finish by evaluating both fitted models on unseen loans and plotting PCR's predictions against observed rates.

1. Use `best_pcr` and `best_poly` to predict interest rates for the test set. Save the PCR predictions as `y_test_hat` and the polynomial predictions as `y_test_poly_hat`. Report the test R^2 for both models using `r2_score()`.
2. Use `scatter()` to plot observed test rates on the horizontal axis and PCR predictions on the vertical axis. Add a dashed line where predicted and observed rates are equal using `axline()`.
3. Briefly describe where the model overpredicts or underpredicts interest rates.

[74]:
```
from sklearn.metrics import r2_score

# TODO: Predict with best_pcr and save the predictions as y_test_hat.

# TODO: Predict with best_poly and save the predictions as y_test_poly_hat.

# TODO: Compute and report the test R² for both models.

# TODO: Plot PCR predictions against observed test rates, with an equality line.
```

Solution.

Both grid searches have refitted their best models on the full training sample. We now predict interest rates for the test sample.

[75]:
```
from sklearn.metrics import r2_score

# Predict outcomes on the test set using the best PCR and polynomial models
y_test_hat = best_pcr.predict(X_test)
y_test_poly_hat = best_poly.predict(X_test)

# Compute the R² scores for both models on the test set
r2_test_pcr = r2_score(y_test, y_test_hat)
r2_test_poly = r2_score(y_test, y_test_poly_hat)

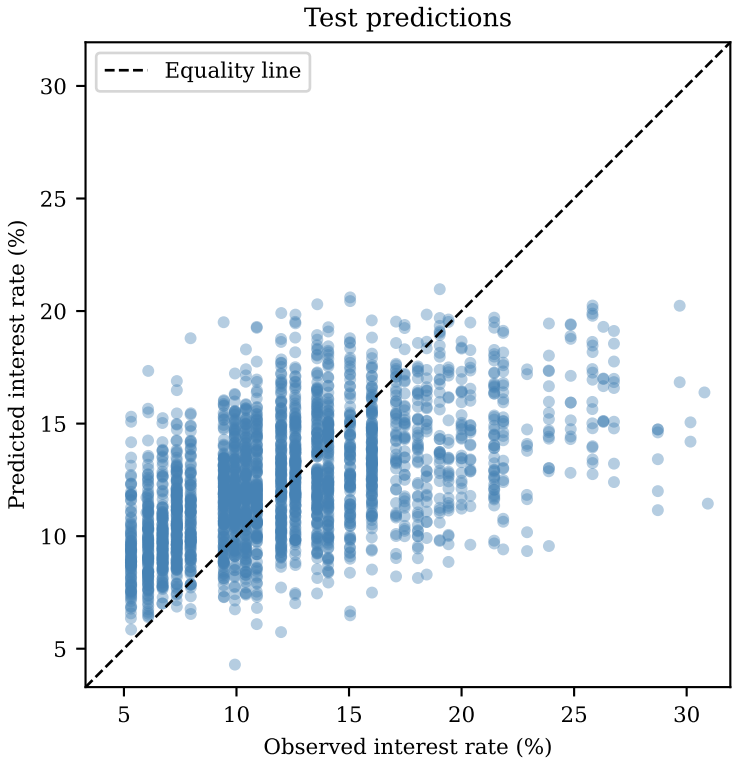
print(f'PCR test R²: {r2_test_pcr:.3f}')
print(f'Polynomial test R²: {r2_test_poly:.3f}')
```

```
PCR test R²: 0.284
Polynomial test R²: 0.288
```

The following code plots the PCR predictions against the observed loan interest rates on the test sample. The dashed line indicates where predicted and observed rates are equal.

[76]:
```
plt.figure(figsize=(4.5, 4), layout='constrained')
# Scatter plot of observed vs predicted interest rates
plt.scatter(y_test, y_test_hat, s=12, alpha=0.4, c='steelblue')
# plot 45-degree equality line
plt.axline((0, 0), slope=1, c='black', ls='--', lw=1, label='Equality line')
plt.gca().set_aspect('equal', adjustable='box')
# Use common limits that include all observed and predicted rates
rate_min = min(y_test.min(), y_test_hat.min()) - 1
rate_max = max(y_test.max(), y_test_hat.max()) + 1
```

```
plt.xlim((rate_min, rate_max))
plt.ylim((rate_min, rate_max))
plt.xlabel('Observed interest rate (%)')
plt.ylabel('Predicted interest rate (%)')
plt.legend()
_ = plt.title('Test predictions')
```



Test R^2 is about 0.284 for PCR and 0.288 for polynomial regression, reversing their small CV difference. This small reversal suggests that the two methods perform similarly, with their ranking depending on the evaluation sample.

The PCR predictions cover a narrower range than observed rates: the model underpredicts many high-rate loans and overpredicts some low-rate loans.